



University of Engineering and Management
Institute of Engineering & Management



INNOVATION & ENTREPRENEURSHIP DEVELOPMENT CELL (IEDC)

Dept. of CSE (Internet of Things, CyberSecurity & Blockchain Technology)

At the IEDC Lab, we are committed to driving technological innovation, fostering entrepreneurial mindsets, and advancing interdisciplinary research. Our focus spans a broad spectrum of next-generation domains, enabling researchers and students to tackle real-world challenges through cutting-edge solutions.

Our Mission

"Empowering Minds, Building Futures: Connecting Education, Engineering, and Management Excellence"

We aim to empower innovators and researchers by providing a collaborative ecosystem that integrates academic rigor with entrepreneurial execution. The IEDC serves as a launchpad for:

- Product innovation and prototyping
- Interdisciplinary project development
- Start-up ideation and incubation support
- Real-world deployment of AI, ML, and IoT solutions

Research Focus Areas

-  **AI in Medical Imaging** – Enhancing diagnostics through deep learning and computer vision.
-  **Brain-Computer Interface (BCI)** – Exploring neurotechnology for cognition, communication, and rehabilitation.
-  **Fintech & Predictive Analytics** – Empowering financial decision-making using machine learning and sentiment analysis.
-  **Smart Agriculture & IoT** – Innovating sustainable farming with real-time monitoring and automation.
-  **Underwater Exploration & Robotics** – Designing intelligent systems for marine research and surveillance.
-  **Nanotechnology & Smart Materials** – Leveraging AI for material property prediction and nanodevice applications.

Advanced Infrastructure

Our lab features state-of-the-art equipment and computational resources for prototyping, experimentation, and scalable model development:

- ⚡ **NVIDIA RTX 5070, A1000, RTX 4090 & RTX 3060 GPUs** – Accelerated training for deep learning and vision models.
- ⚡ **RMS Superspec 32-Channel EEG Machine** – High-resolution EEG acquisition for BCI research.
- ⚡ **NAS Server** – High-speed, secure data storage for large-scale experiments.
- ⚡ **NVIDIA Jetson Nano & Orin** – Real-time edge AI computing for smart devices.
- ⚡ **Industry-Grade Prototyping Tools** – 3D printer, Soldering, PCB design, and embedded hardware integration.
- ⚡ **IoT Platforms (Arduino, Raspberry Pi 5, ESP32)** – Practical development for smart, connected systems.

Collaborate With Us

We welcome partnerships with industry, academia, and government agencies to jointly explore, co-develop, and deploy impactful technology solutions.

👉 Join us as we build the future of intelligent systems and transform ideas into reality!

Contact with us,

Email id: iedciotlab@gmail.com

LinkedIn Page: https://www.linkedin.com/posts/uem-innovation-and-entrepreneurship-development-cell_ai-deeplearning-cognitiveneuroscience-activity-7337555871637323777-mspl?utm_source=share&utm_medium=member_desktop&rcm=ACoAACIZcYoBwqxYEmre6WL2zXlm-I5XAdIYAb4

RESEARCH HIGHLIGHTS (2024–2025)

Applied AI in Healthcare and Biomedical Engineering

1. Predictive Modelling for Chronic Kidney Disease

 *IEEE IEMENTech 2025*

 *Authors:* Apurba Nandi, Priyanka Paul, Avik Kumar Das, Arijeet Ghosh, Anjali Singh, Aayush Gupta

 DOI: [10.1109/IEMENTech65115.2025.10959403](https://doi.org/10.1109/IEMENTech65115.2025.10959403)

2. Judicious Use of Anti-VEGF in Diabetic Macular Edema

 *Scientific Reports, Nature Springer (Q1)*

 *Authors:* Anwesha Mondal, Apurba Nandi, Subhasish Pramanik & Lakshmi Kanta Mondal (UEM, RIO, IISc collaboration)

 <https://doi.org/10.1038/s41598-025-87290-3>

3. Alzheimer's Disease Detection via ML

 *Springer IESIA 2025 (Accepted for Publication)*

 *Authors:* Faishal Abrar, Apurba Nandi, Sangita Dutta, Sandip Mandal

4. Gestational Diabetes Prediction using Meta Learning & XAI

 *Springer ICACCP 2025 (Accepted for Publication)*

 *Authors:* Yashraj Sharma, Suvajit Majhi, Rekharani Mahanta, Avik Kumar Das, Apurba Nandi

5. Design IoT and Blockchain-Driven System for Real-Time Organ Donation Management

 *Springer ICDMIS LNNS 2025 (Accepted for Publication)*

 *Authors:* Siddhartha Roy, Avik Kumar Das, Sandip Mandal, Subhajit Ghosh

6. Brain Tumour Diagnosis with Ensemble Deep Learning Technique

 *Springer AISC 2025 (Accepted for Publication)*

 *Authors:* Apurba Nandi, Tista Mukherjee, Soumika Das, Nitu Saha, Sayantani Das, Sandip Mandal

Tourism & Sustainability

7. Meta-Deep Learning for Fuel Analytics

 *Springer Book Chapter: "Internet of Vehicles" (2025) (Accepted for Publication)*

 *Authors:* Apurba Nandi, Sandip Mandal, Sayantani Das, Arijeet Ghosh, Avik Kumar Das

8. Bishnupur Architecture Analysis & Tourism Enhancement with Deep Learning

 *IEEE INDISCON 2025 (Accepted for Publication)*

 *Authors:* Apurba Nandi, Avik Kumar Das, Tuheena Bose, Divyanshi Srivastava, Sandip Mandal, Madhurya Chowdhury

AI-Driven Intelligent Systems & Platforms

9. Student-Alumni Networking via DQN & Graph Intelligence

 *IEEE INCIP 2025*

 *Authors:* N. Kundu, G. Mandal, A. Bhandari, U. Bhaskar, Apurba Nandi

 DOI: [10.1109/INCIP64058.2025.11019749](https://doi.org/10.1109/INCIP64058.2025.11019749)

10. Job Recommendation Using Real-Time Skills and Trends

 *IEEE CIACON 2025 (Accepted for Publication)*

 *Authors:* Apurba Nandi, Avik Kumar Das, Hritajit Sur, Sankalpa Dutta, Arijeet Das, Shuvam Podder

11. Firewall Defense via Reinforcement Learning

 *Bentham Science Book Chapter 2025 (Accepted for Publication)*

 *Authors:* Apurba Nandi, Uddipan Ghosh, Sangita Dutta, Sandip Mandal, Avik Kumar Das, Arijeet Ghosh

12. Graph-Based Student-Alumni Engagement

 *Springer RAISAA 2024 (Accepted for Publication)*

 *Authors:* Danisha Basu, Hritajit Sur, Dwaipyan Ghosh, Aritra Das, Apurba Nandi, Arijeet Ghosh

13. Overlapping Pairwise Chunking and Merging for Long Context Summarization in LLMs

 IEEE SCRS AIC 2025 (Accepted for Publication)

 Authors: Avik Kumar Das, Subhajit Ghosh

14. Tracking the Tides of Toxicity: Temporal Shifts in Public Emotion and Discourse During the COVID-19 Crisis

 IEEE Conference International Conference on Advances in Computing Research on Science Engineering and Technology (ACROSET 2025), in association with #IEEE_MP_Section,

 Authors: Uddipan Ghosh, Sangita Dutta, Indrajit Mondal, Apurba Nandi, Avik Kumar Das

Financial Forecasting & Economic Intelligence

15. Stock Forecasting Using Bi-LSTM-GRU Hybrid Model

 Taylor & Francis IRTM 2024 (Accepted for Publication)

 Authors: Shaoni Banerjee, Hansa Ray, Shubhanshu Tiwari, Khizra Aslam, Apurba Nandi, Avik Kumar Das

16. Fuzzy AI for Riverbank Health Monitoring

 Springer ICDMIS 2025 (Accepted for Publication)

 Authors: Apurba Nandi, Anjali Singh, Aasish Shrestha, Tanisha Pan, Anwesha Das, Avik Kumar Das

17. Gold Stock Forecasting with FinBERT & Federated Learning

 Springer ICDMIS 2025 (Accepted for Publication)

 Authors: Apurba Nandi, Pratyay Ghosh, Kuntal Roy, Puspak Samanta, Diptadip Choudhury, Anwesha Ghosh

Neuroscience & Cognitive AI

18. Cognitive Load Estimation Using Deep Neural Networks

 Published in: Cognitive Neurodynamics Journal (Springer Nature)

 Impact Factor: 3.10 | SCIE Indexed

 Authors: Shaoni Banerjee, Priyanka Paul, Apurba Nandi, Avik Kumar Das, Arijeet Ghosh

 Highlights:

- Presents a multilayer deep neural network framework for hemodynamic assessment of cognitive load during problem-solving tasks.
- Integrates cognitive neuroscience with AI for mental workload estimation in real-time environments.
- Aimed at enhancing adaptive learning systems, brain-computer interfaces.

19. AI-Driven Brain-Machine Interface system for Effortless Communication with physically Challenged Individuals

 IEEE INDISCON 2025 (Accepted for Publication)

 Authors: Shibagni Bhattacharjee, Debajyoti Chatterjee, Suchanda Chatterjee

Agriculture

20. Design of IOT Enabled Smart Agriculture System Using Automated Vehicle

 IEEE INDISCON 2025 (Accepted for Publication)

 Authors: Arka Mahajan, Ishika Halder, Srishti Srinjan, Debajyoti Chatterjee, Suchanda Chatterjee

21. Attention-Enhanced Deep Learning Framework with Edge-Aware Preprocessing for Ground-Based Cloud Classification

 Authors: Apurba Nandi, SAYANTANI DAS, Nitu Saha, Tanisha Pan, Sukriti Sikdar, Anwesha Sinha, Tamoshree Dey

 Conference: 2nd International Conference on Data Mining and Information Security (ICDMIS_2025), supported by Springer (Accepted for Publication)

22. Phytosense: A Lightweight System for Real-Time Plant Disease Detection

 Authors: Sudipta Dey, Akash Mondal, Arijeet Ghosh, Sreya Ghosh, Apurba Nandi, Kiran Ghosh

 Conference: 3rd International Conference on Data Analytics and Insights, Springer (ICDAI 2025) (Accepted for Publication)

Secure Systems & Data Integrity

23. Blockchain for Secure E-Voting Systems

 Title: Ensuring Security and Transparency in E-Voting System through Blockchain Technology

 Conference: International Conference on Data Mining and Information Security (ICDMIS-2024), Springer (Accepted for Publication)

 Author: Siddhartha Roy

24. Transforming Crowdfunding with Blockchain

 Title: Transforming Crowdfunding with Blockchain: Unlocking the Future of Fundraising

 Conference: 8th International Conference on Innovative Computing and Communication (ICICC-2025) Elsevier SSRN (Accepted for Publication)

 Author: Dr. Siddhartha Roy, Adarsh Giri, Yash Kumar Giri, Akash Mondal, Md Asid Khan

25. Social Media, Polarization, and the Erosion of Informed Voting: The Impact of Echo Chambers

 Conference: Innovative Strategies in Intelligent Computing and Communication (ICOICC 2025) (Accepted for Publication)

 Author: Sangita Dutta

26. Transparency and Trust for Religious Organizations

 *Title: Developing a Blockchain-Based Framework to Foster Transparency and Trust for Religious Organizations and NGOs*

 *Conference: ICDMIS 2025 Springer (Accepted for Publication)*

 *Author: Dr. Siddhartha Roy, Adarsh Giri, Yash Kumar Giri, Akash Mondal, Md Asid khan*

Applied Physics & NanoTech

27. Photoluminescence Prediction in Silicon Nanostructures

 *Title: Ensemble Based Prediction Analytics Model for Photoluminescence Intensity Decay in Silicon Nanostructures*

 *Conference: International Conference on Data Management, Analytics and Innovation (ICDMAI 2025), Springer (Accepted for Publication)*

 *Author: Siddhartha Roy, Susmita Biswas*

28. Predictive Modeling of Photoluminescence spectrum

 *Title: Optimization and Comparison of Machine Learning Models for Predictive Modeling of Photoluminescence spectrum of freestanding colloidal Silicene ink: A Multi-Factor Performance Evaluation Framework*

 *Conference: AdComsys 2025 Springer (Accepted for Publication)*

 *Author: Susmita Biswas, Siddhartha Roy, Rangoon Sarkar and Raisa Santra*

Book Chapters

29. Explainable AI for Fuel and Energy Consumption

Springer Book: Internet of Vehicles (Accepted for Publication)

Authors: Apurba Nandi, Sandip Mandal, Sayantani Das, Arijeet Ghosh, Avik Kumar Das

30. Micro Intelligent System Technology for Adaptive Firewall Defense

Bentham Science (2025) (Accepted for Publication)

Authors: Apurba Nandi, Uddipan Ghosh, Sangita Dutta, Sandip Mandal, Avik Kumar Das, Arijeet Ghosh

31. ***Blockchain Technology in the Global Supply Chain: A Theoretical Overview and Security Issues***

Contemporary Issues in International Trade, Emerald Publishing Limited, May 2024

DOI: <https://doi.org/10.1108/978-1-83797-320-020241012>

Authors: Dr. Siddhartha Roy

32. Dr. Siddhartha Roy **Editor of Cutting-Edge Bioinformatics Volume Edited Book: *Exploring Cutting-Edge Intelligent Applications of Bioinformatics: A Practical Perspective*** (Accepted for Publication)

CRC Press, Taylor & Francis Group, USA (2024)

33. Dr. Siddhartha Roy **Editor of Volume on Deep Neuro-Fuzzy & Explainable AI, *Intelligent Systems: Emerging Trends and Challenges in Deep Neuro-Fuzzy Models and Explainable AI***

Apple Academic Press (CRC Press, Taylor & Francis Group, USA) (2024) (Accepted for Publication)

34. LSTM-Based E-commerce Sales Prediction

Book Chapter Title: *Enhancing E-commerce Sales Prediction: Overcoming Data and Complexity Challenges with LSTM Models (Accepted for Publication)*

CRC Press, Taylor & Francis Group, USA: *Intelligent Systems: Emerging Trends and Challenges in Deep Neuro Fuzzy Models and Explainable Artificial Intelligence*

Authors: Sangita Dutta, Dr. Siddhartha Roy

35. Interdisciplinary Advances in AI

Book Nova Science Publisher: *Intelligent Systems and Emerging Technologies: Interdisciplinary Advances in AI, Engineering, and Computational Modeling (Accepted for Publication)*

Authors: Dr. Siddhartha Roy, Dr. Malay Gangopadhyaya, Dr. Sandip Mandal

36. Tele-Oncology in Bioinformatics

Chapter Title: *An Intervention of Tele-Oncology in Palliative Care: A Critical Review and Analysis of Remote Monitoring, Cancer Diagnosis, and Treatment (Accepted for Publication)*

Apple Academic Press (CRC Press): *Intelligent Applications of Bioinformatics: A Practical Approach*

Authors: Partha Sarker, Dr. Siddhartha Roy

37. AI in Drug Discovery and Telemedicine

Chapter Title: *Machine Learning Transformations in Drug Discovery: A Paradigm Shift in Development Strategies (Accepted for Publication)*

De Gruyter Publishers (Scopus Indexed): *Drug Discovery and Telemedicine Through Artificial Intelligence, Computer Vision, and IoT*

Author: Dr. Siddhartha Roy

Awards & Recognitions

1. 🏆 Aegis Graham Bell Award Finalist (Feb 2025), Supported by Govt. of India

Project: AquaSmartIQ – A smart Aquaculture monitoring platform using IoT and AI

🔍 Team: Apurba Nandi, Avik Kumar Das, Sandip Mandal, Arijeet Ghosh



Image 1: University of Engineering & Management (UEM), was recognized as a finalist under the 'Innovation in RPA' category and **Dr. Sandip Mandal**, Head of the Department, CSE (IoT, CS & BT), received the award in the esteemed presence of **Hon'ble Minister Shri Jitin Prasada**, Union Minister of State for Commerce and Industry, and Electronics and Information Technology, Government of India.



Image 2: The award was graciously conferred upon our esteemed Vice Chancellor Sir, **Dr. Sajal Das Gupta**, accompanied by team members **Avik Kumar Das** and **Apurba Nandi** in recognition of their contribution.